PassFrame
tuition

Ratio & Proportion

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

A) Easy: Share £20 in the ratio 1:4.
Method & answer: Total parts = 5. Each part = £4. Shares: £4 and £16.

B) Standard: A recipe uses flour and sugar in the ratio 3:2. There are 300g flour. How much sugar?
Method & answer: 1 part = 100g (300 ÷ 3). Sugar = 2 × 100 = 200g.

C) Advanced: Two numbers are in the ratio 5:3 and their sum is 96. Find the numbers.
Method & answer: 8 parts = 96, so 1 part = 12. Numbers: 5×12=60 and 3×12=36.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Simplify the ratio 4:8.
2. Simplify the ratio 10:15.
3. Simplify the ratio 6:9.
4. Share 12 sweets in the ratio 1:2.
5. Share 20 marbles in the ratio 1:4.
6. Share 18 in the ratio 1:2.
7. Simplify the ratio 12:16.
8. Write the ratio of 3 apples to 9 oranges in simplest form.
9. Share 30 in the ratio 2:3.
10. Simplify the ratio 8:20.
11. Share 24 in the ratio 1:3.
12. Write the ratio 15:25 in simplest form.
13. Share 40 in the ratio 3:5.
14. Simplify the ratio 9:12.
15. Share 21 in the ratio 2:5.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Share £96 in the ratio 3:5.
2. Share 132kg in the ratio 2:3:7.
3. A ratio is 4:7. If the smaller amount is 28, what is the larger?
4. A ratio is 3:8. If the larger amount is 96, what is the smaller?
5. Share 156 in the ratio 3:4:5.
6. If 6 pens cost £4.50, how much do 15 pens cost?
7. If 5 books cost £22, how much do 8 books cost?
8. A map scale is 1:250. A distance of 6cm on the map is how many metres in real life?
9. For every 3 boys in a class there are 5 girls. If there are 18 boys, how many girls?
10. Simplify the ratio 42:126.
11. Share 117 in the ratio 4:9.
12. A ratio is 7:9. If the total is 144, find both parts.
13. Write 45p to £3 as a ratio in its simplest form.
14. If 8kg of oranges costs £14, how much do 12kg cost?
15. Share 210 in the ratio 4:3:7.

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A recipe for 4 people needs 300g of rice. How much rice is needed for 10 people?

2. Two numbers are in the ratio 3:7 and differ by 20. Find the two numbers.

3. A fruit drink is made from orange and apple juice in the ratio 5:3. How many ml of orange juice are in 800ml of the drink?

4. In a school, the ratio of teachers to pupils is 1:18. If there are 36 teachers, how many pupils are there?

5. A sum of money is shared between Amy, Ben and Cal in the ratio 2:3:5. Cal receives £45 more than Amy. How much does Ben receive?

6. Paint is mixed in the ratio blue:white = 2:5. How much white paint is needed for 60ml of blue paint?

7. A car travels 180 miles using 6 gallons of fuel. At the same rate, how many miles can it travel on 10 gallons?

8. The ratio of cats to dogs at a shelter is 3:4. If there are 63 animals in total, how many are dogs?

9. A recipe requires flour, sugar and butter in the ratio 6:3:2. If 180g of flour is used, how much sugar and butter combined is used?

10. Two numbers are in ratio 4:9. Their sum is 130. Find the difference between the numbers.

11. A photo measuring 4cm by 6cm is enlarged in the ratio 1:3. What are the new dimensions?

12. 8 workers can build a wall in 15 days. How many days would it take 12 workers, working at the same rate?

13. A model car is built to a scale of 1:24. If the model is 8cm long, how long is the real car in metres?

14. In a bag of counters, red:blue:green = 3:4:5. There are 15 red counters. How many counters are there in total?

15. A drink recipe uses squash and water in ratio 1:6. If a jug holds 2.1 litres in total, how much squash is used?